

MATHEMATICS CLASS – X WBBSE 2017

1. Choose the correct option in each case from the following questions: (1X6=6)
- If the ratio of principal and yearly amount be in the ratio 25:28, then the yearly rate of interest is_____
(a) 3% (b) 12% (c) $10\frac{5}{7}\%$ (d) 8%
 - Under what condition one root of the quadratic equation $ax^2 + bx + c = 0$ is zero?
(a) $a=0$ (b) $b=0$ (c) $c=0$ (d) None
 - The number of common tangents of two circles when they do not touch or intersect each other is_____.
(a) 2 (b) 1 (c) 3 (d) 4
 - If $\sin \theta = \cos \theta$, then value of 2θ will be _____
(a) 30° (b) 60° (c) 45° (d) 90°
 - If each of radius of base and height of a cone be doubled, then the volume of it will be _____ of previous one.
(a) 3 times (b) 4 times (c) 6 times (d) 8 times
 - The median of the numbers 2, 8, 2, 3, 8, 5, 9, 5, 6 is _____.
(a) 8 (b) 6.5 (c) 5.5 (d) 5
2. Fill in the blanks (any Five):- (1X5=5)
- At same rate per cent per annum, the simple interest and compound interest of same principal are same in _____year.
 - If in a quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$) $b^2 = 4ac$, then the roots of the equation will be real and _____.
 - If the length of the sides of two triangles are in proportion, then two triangles are _____.
 - If $\cos^2 \theta - \sin^2 \theta = \frac{1}{x}$, $x > 1$ then $\cos^4 \theta - \sin^4 \theta =$ _____.
 - The number of plane surface of a solid hemisphere is _____.
 - If the mean of $x_1, x_2, x_3, \dots, x_n$ be $\bar{x}$, what is the mean of $Kx_1, Kx_2, Kx_3, \dots, Kx_n$ where ($K \neq 0$) is _____.
3. Write True or False (any Five):- (1X5=5)
- A starts a business with Rs. 10000 and B gives Rs. 20000 after 6 months. At the end of the year their profit will be equal.
 - If $x = 2 + \sqrt{3}$, then the value of $x + \frac{1}{x} = 2\sqrt{3}$.
 - If two circles of radii 7 cm and 3 cm touch each other externally, the the distance between their centres will be 4 cm.
 - If $0^\circ < \theta < 90^\circ$, then $\sin \theta > \sin^2 \theta$.

MATHEMATICS CLASS – X WBBSE 2017

- v. If the total surface area of a hemisphere be 36π sq. cm., then its radius will be 3 cm.
- vi. If the perpendicular drawn on x axis from the point of intersection of both ogives, the abscissa of the point of intersection of the perpendicular with the x axis will be the median.
4. Answer the following questions (any ten):- (2X10=20)
- i. A sum of money is doubled in 8 years at $r\%$ rate of compound interest per annum. At the same rate in how many years it will be four times of the sum?
- ii. A invests $1\frac{1}{2}$ times more than B invests in a business. At the end of the year B receives Rs 1500 as profit. How much profit A will get at the end of that year?
- iii. Without solving, find the values of 'p' for which the equation $x^2 + (p - 3)x + p = 0$ has real and equal roots.
- iv. If $x \propto yz$ and $y \propto zx$, then show that z is a non-zero constant.
- v. The perimeter of two similar triangles are 20 cm and 16 cm respectively. If the length of one side of the first triangle is 9 cm, then find the length of the corresponding side of the second triangle.
- vi. In triangle ABC, $\angle ABC = 90^\circ$, $AB = 5$ cm and $BC = 12$ cm, find the length of the circum radius of the triangle ABC.
- vii. In triangle ABC, $\angle ABC = 90^\circ$, $AB = (2a - 1)$ cm, $AC = 2\sqrt{2}$ cm, $BC = (2a + 1)$ cm, then write down the value of $\angle BAC$.
- viii. If $x = a \sec \theta$, $y = b \tan \theta$, then find the relation between x and y free from θ .
- ix. If $\tan(\theta + 15^\circ) = \sqrt{3}$, then find the value of $\sin \theta + \cos \theta$.
- x. The diameter of one sphere is double the diameter of another sphere. If the value of total surface area of larger sphere is equal to the volume of the smaller sphere, then find the radius of the smaller sphere.
- xi. If the number of surfaces of a cuboid is x, the number of edges is y, the number of vertices is z and the number of diagonals is P, then find the value of $x - y + z + P$.
- xii. If 11, 12, 14, x-2, x+4, x+9, 32, 38, 47 are arranged in ascending order and their median is 24, find x.
5. Answer any one question:- (1X5=5)
- i. The difference between simple interest and compound interest for 2 years of a sum of money becomes Rs. 80 at 4% interest per annum. Calculate the sum of money.
- ii. A, B, C starts a business jointly investing Rs. 180000 in together. A gives Rs. 20000 more than that of B and B gives Rs. 20000 more than that of C. Distribute the profit of Rs. 10800 among them.

MATHEMATICS CLASS – X WBBSE 2017

6. Solve any one question:- (1X3=3)

- i. $\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}$, $[x \neq 0, -(a+b)]$
- ii. If 5 times of a positive whole number is less by 3 than twice of its square, then find the number?

7. Answer any one question:- (1X3=3)

- i. Simplify $\frac{1}{\sqrt{2}+\sqrt{3}} - \frac{\sqrt{3}+1}{2+\sqrt{3}} + \frac{\sqrt{2}+1}{3+2\sqrt{2}}$.
- ii. The total expenses of a hostel are partly constant and partly vary directly as the number of boarders. When the numbers of boarders are 120 and 100 the total expenses are Rs. 2000 and Rs. 1700 respectively. What will be the numbers of boarders when the total expenses is Rs. 1800?

8. Answer any one question:- (1X3=3)

- i. If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b}$, then prove that each ratio is equal to $\frac{1}{2}$ or -1.
- ii. If $(b+c-a)x = (c+a-b)y = (a+b-c)z = 2$, then prove that
$$\left(\frac{1}{x} + \frac{1}{y}\right)\left(\frac{1}{y} + \frac{1}{z}\right)\left(\frac{1}{z} + \frac{1}{x}\right) = abc.$$

9. Answer any one question:- (1X5=5)

- i. If in a triangle, the area of the square drawn on one side equal to the sum of the areas of squares drawn on other sides, then prove that the angle opposite to the first side will be a right angle.
- ii. If two tangents are drawn to a circle from a point outside it, then the line segment joining the point of contact and the exterior points are equal.

10. Answer any one question:- (1X3=3)

- i. Prove that the quadrilateral formed by the internal bisector of the four angles of a quadrilateral is cyclic.
- ii. O is the circum-centre of triangle ABC, $OD \perp BC$ then prove that $\angle BOD = \angle BAC$.

11. Answer any one question:- (1X5=5)

- i. Draw an equilateral triangle of side 6 cm and draw the incircle of the triangle. (only traces of construction required)
- ii. Construct a rectangle with sides 8 cm and 6 cm and construct a square equal in area to that of rectangle. (only traces of construction required)

12. Answer any two questions:- (3X2=6)

- i. If the measures of the three angles of a quadrilateral are $\frac{\pi}{3}$, $\frac{5\pi}{6}$ and 90° , then determine the fourth angle in sexagesimal and circular measure.

MATHEMATICS CLASS – X WBBSE 2017

- ii. If $\frac{\sin \theta}{x} = \frac{\cos \theta}{y}$, then prove that $\sin \theta - \cos \theta = \frac{x - y}{\sqrt{x^2 + y^2}}$.
- iii. If $\tan 9^\circ = \frac{a}{b}$, then show that $\frac{\sec^2 81^\circ}{1 + \cot^2 81^\circ} = \frac{b^2}{a^2}$.

13. Answer any one question:-**(1X5=5)**

- The distance between two pillars is 150 m. Height of one is thrice of other. From the midpoint of the line segment joining the foot of the pillars, the angle of elevation of the top of the pillars are complementary to each other, then find the height of the shortest pillar.
- If the angle of the depression from a lighthouse of two ships situated in the same line with the lighthouse are 60° and 45° & if the nearer ship is 150 m away from the lighthouse, then find the distance of the other ship from the lighthouse.

14. Answer any two questions:-**(4X2=8)**

- Determine the volume of a solid right circular cone which can be made from a solid cube of 4.3 dcm edge length by wasting minimum quantity of wood.
- A hemispherical bowl with radius of 9 cm is to completely filled with water. How many cylindrical bottles of diameter 3 cm and height 4 cm can be filled up with the water in the bowl?
- Area of the base of a closed cylindrical tank is 616 sq. meter and height is 21 meter. Find the total surface area of the tank.

15. Answer any two questions:-**(4X2=8)**

- If the median of the following data is 32, find the values of X and Y when the sum of the frequencies is 100:

Class	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	10	X	25	30	Y	10

- Find the mode from the following distribution table:

Class	0-5	5-10	10-15	15-20	20-25	25-30	30-35
Cumulative Frequency	5	12	18	28	17	12	8

- By preparing a cumulative frequency (greater than type) table from the following data, draw an ogive in a graph paper.

Class	0-5	5-10	10-15	15-20	20-25	25-30
Frequency	4	10	15	8	3	5

[Additional Questions for External Candidates]**16. (a) Answer any three questions:****(2X3=6)**

MATHEMATICS CLASS – X WBBSE 2017

- i. If $x \propto y$, $y \propto z$, $z \propto x$, then find the relation between the constants of variations.
- ii. If in the case of compound interest, the rate of interest in 1st, 2nd and 3rd year are $r_1\%$, $r_2\%$, $r_3\%$, respectively, then find the amount after 3 years for Rs. P.
- iii. If $\sqrt{2} \sin(2x + 5^\circ) = \tan 45^\circ$, find the value of $\sec 3x$.
- iv. What is the ratio of the surface area of a solid sphere and a solid hemisphere of equal radius?

(b) Answer any four questions:-

(1X4=4)

- i. If $(a + 2b) : (3a - 2b) = 9 : 13$, then find the value of $a : b$?
- ii. What should be subtracted from $\sqrt{72}$ to get $\sqrt{32}$?
- iii. What is the length of the tangent drawn from a point at a distance of 13 cm apart from the centre of the circle whose radius is 5 cm?
- iv. Find the mean proportional between 3 and 12.
- v. Justify that the roots of the equation $4x^2 - 4x + 1 = 0$ are real and equal.
